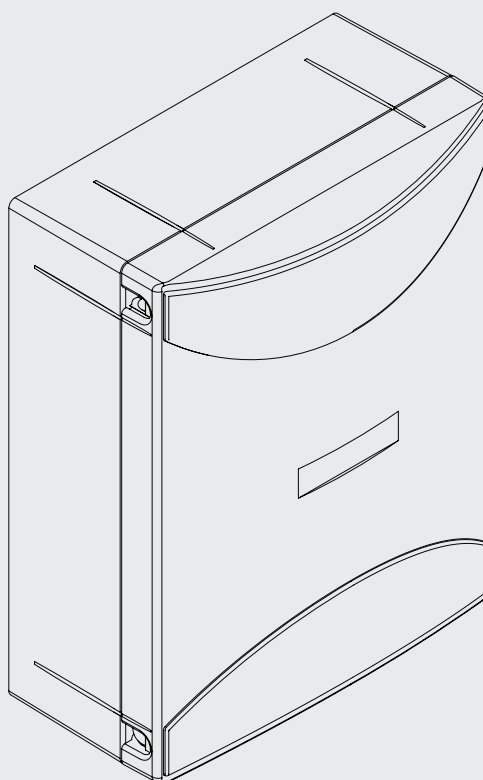
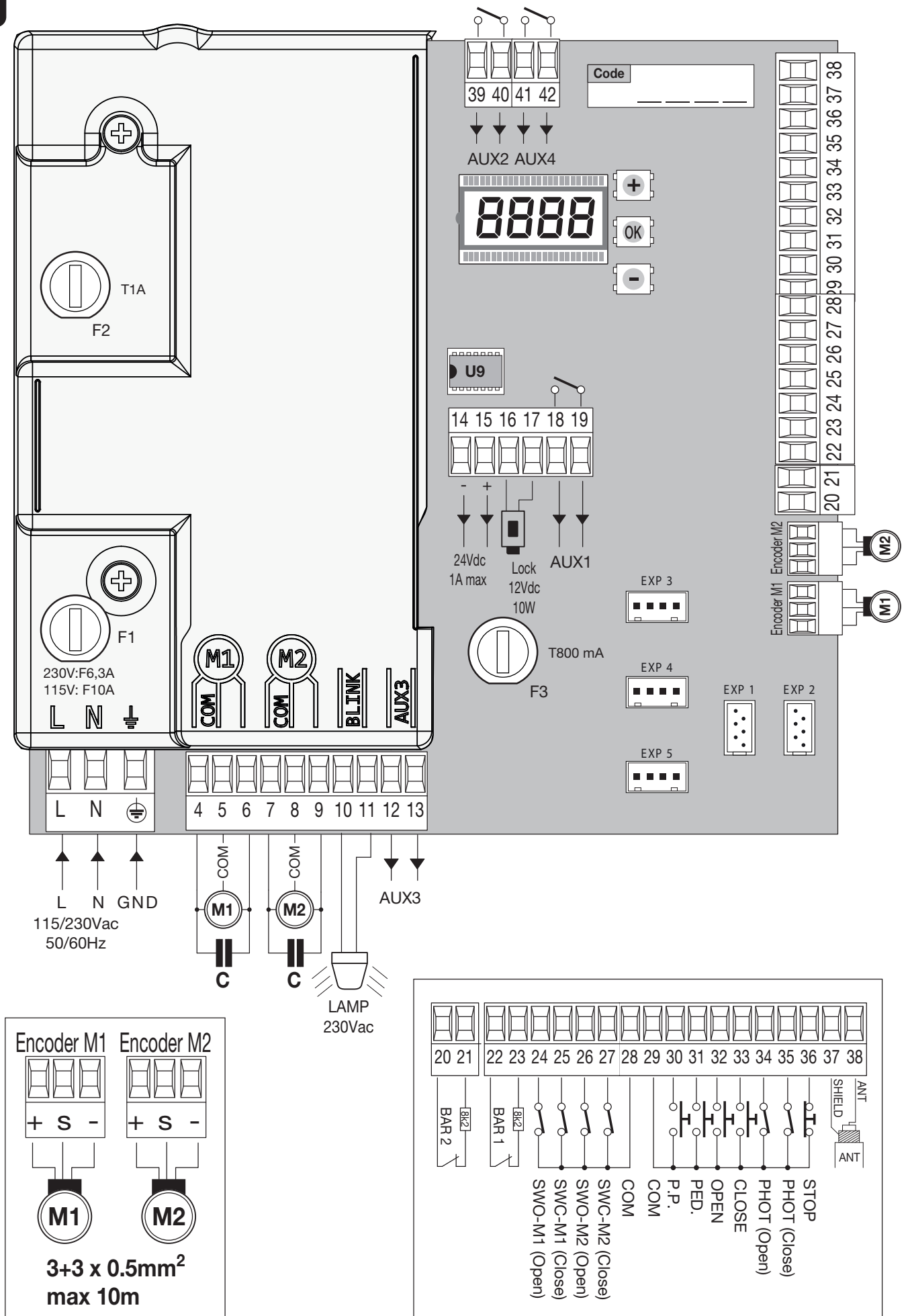


BRAINY PLUS

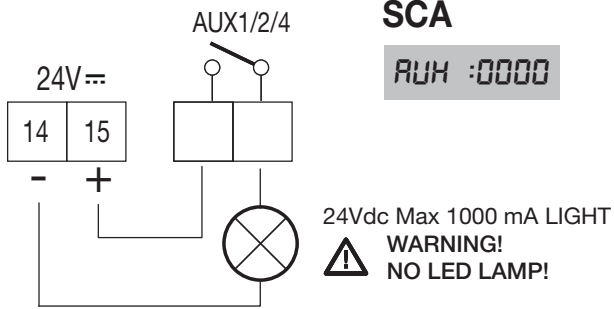


BENINCA[®]
TECHNOLOGY TO OPEN



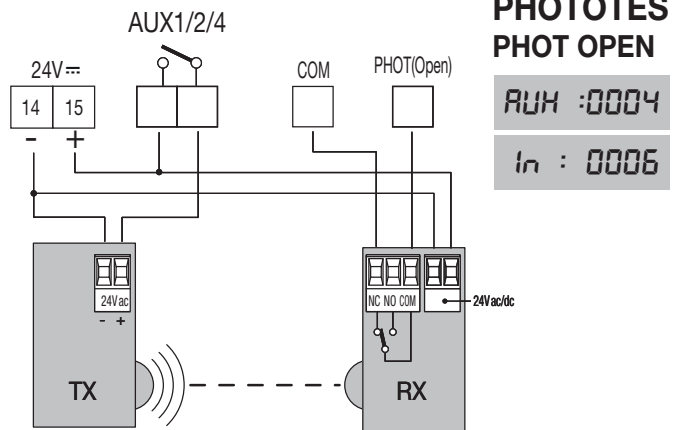
AUX 1 / AUX 2 / AUX 4

SCA



AUX1/2/4

AUX :0000

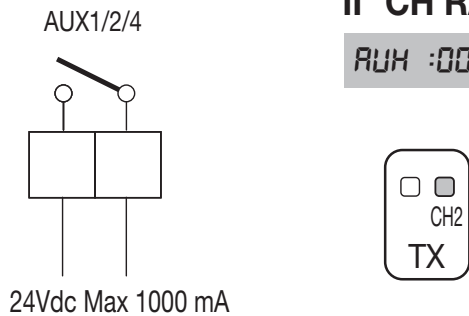
PHOTOTEST
PHOT OPEN

AUX1/2/4

AUX :0004

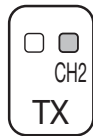
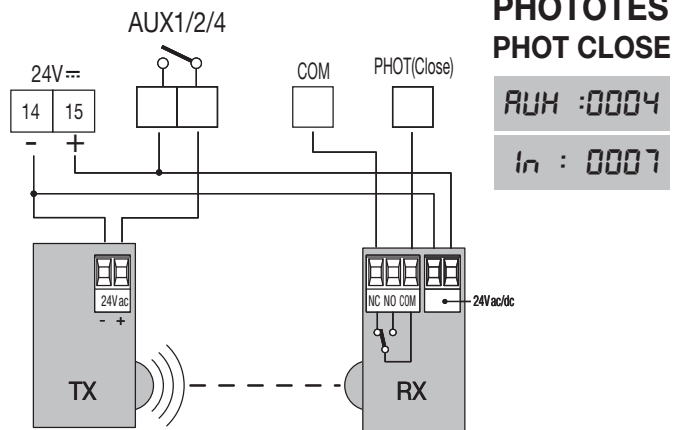
In : 0006

II° CH RADIO



AUX1/2/4

AUX :0001

PHOTOTEST
PHOT CLOSE

AUX1/2/4

AUX :0004

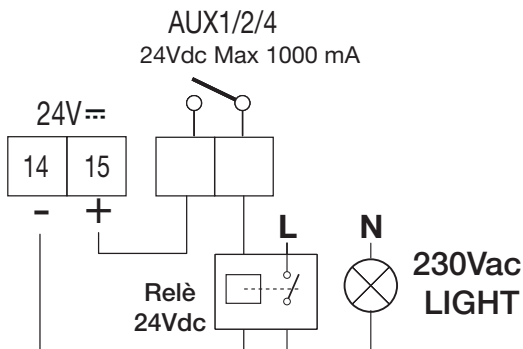
In : 0007

Service Light
(tL 5 Time)

AUX :0002

Zone Light

AUX :0003



AUX1/2/4

24Vdc Max 1000 mA

24Vdc

14 15

-

+

Relè

24Vdc

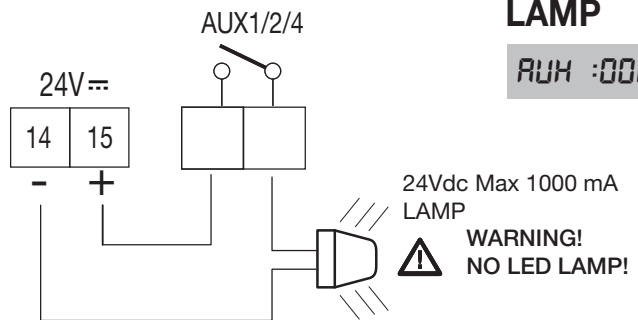
L

N

230Vac
LIGHT

LAMP

AUX :0005



AUX1/2/4

24Vdc

14 15

-

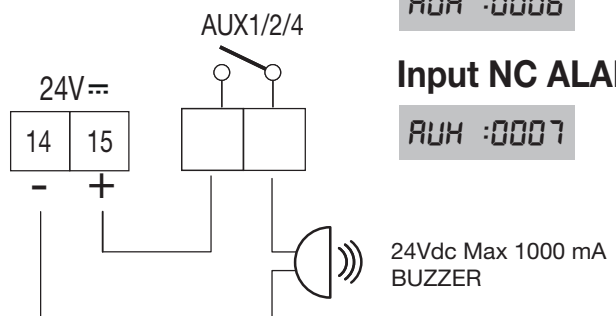
+

Open Gate ALARM

AUX :0006

Input NC ALARM

AUX :0007



AUX1/2/4

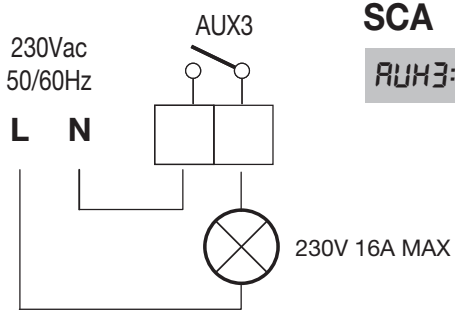
24Vdc

14 15

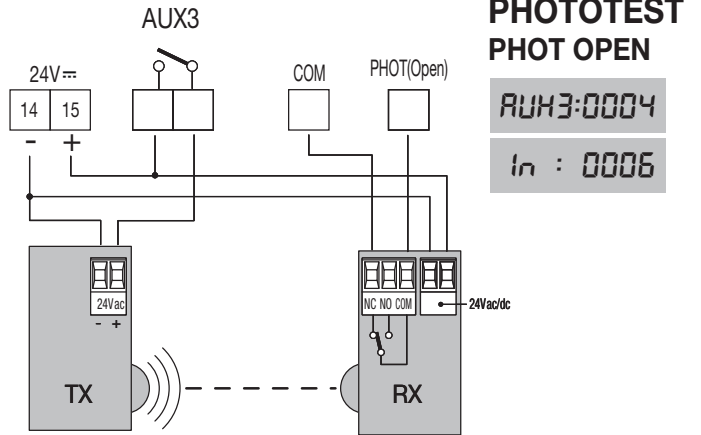
-

+

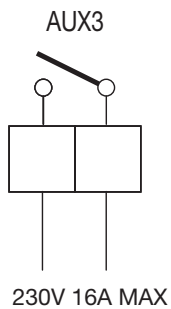
AUX 3



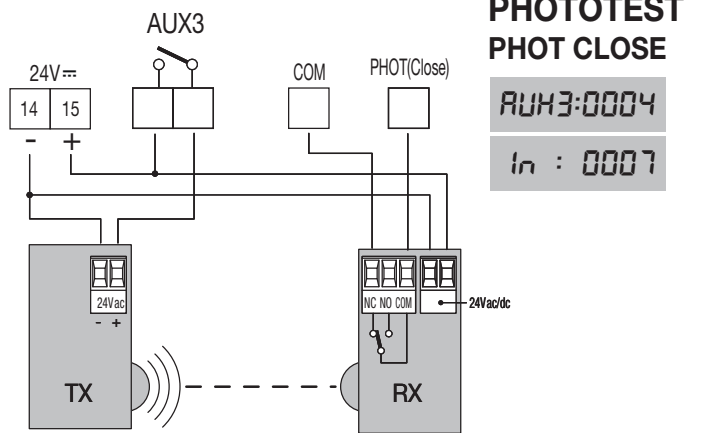
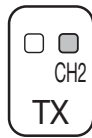
SCA
AUX3:0000



AUX3:0004
In : 0006



II° CH RADIO
AUX3:0001



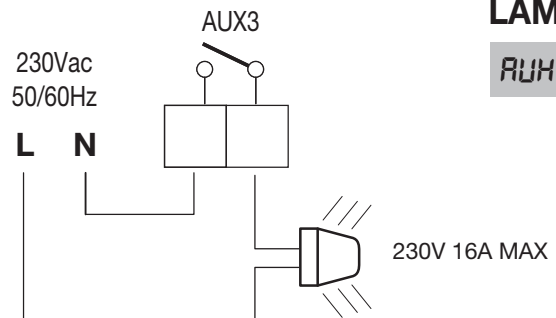
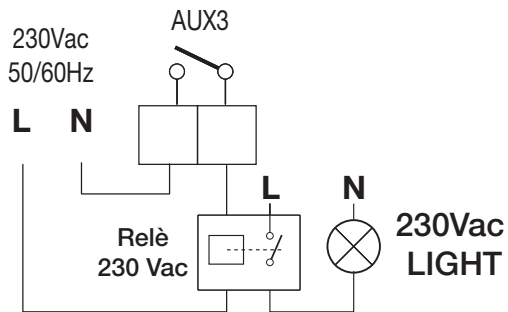
AUX3:0004
In : 0007

Service Light
(t-L 5 Time)

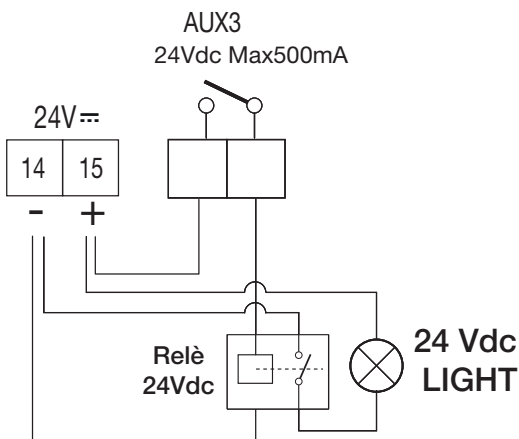
AUX3:0002

Zone Light

AUX3:0003



LAMP
AUX3:0005

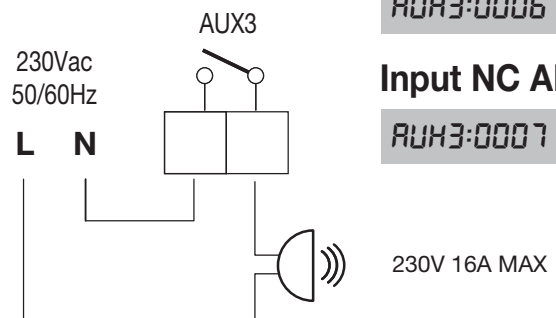


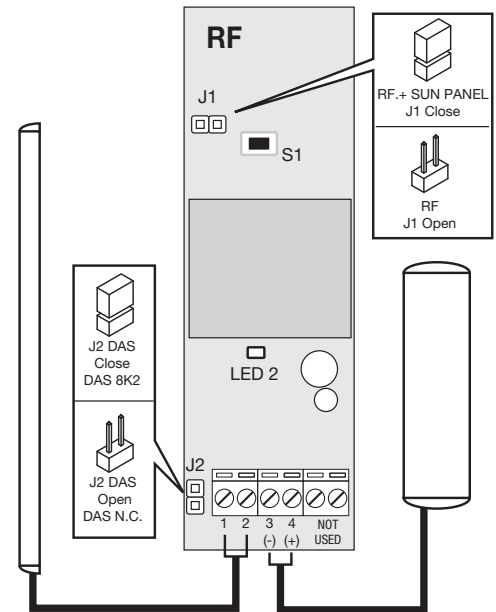
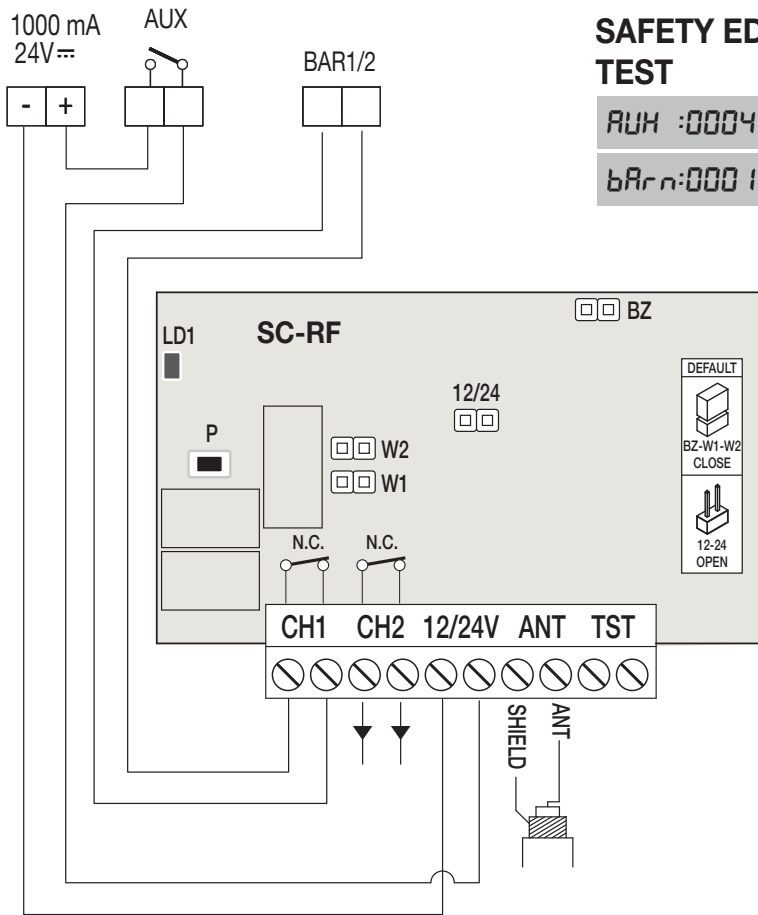
Open Gate ALARM

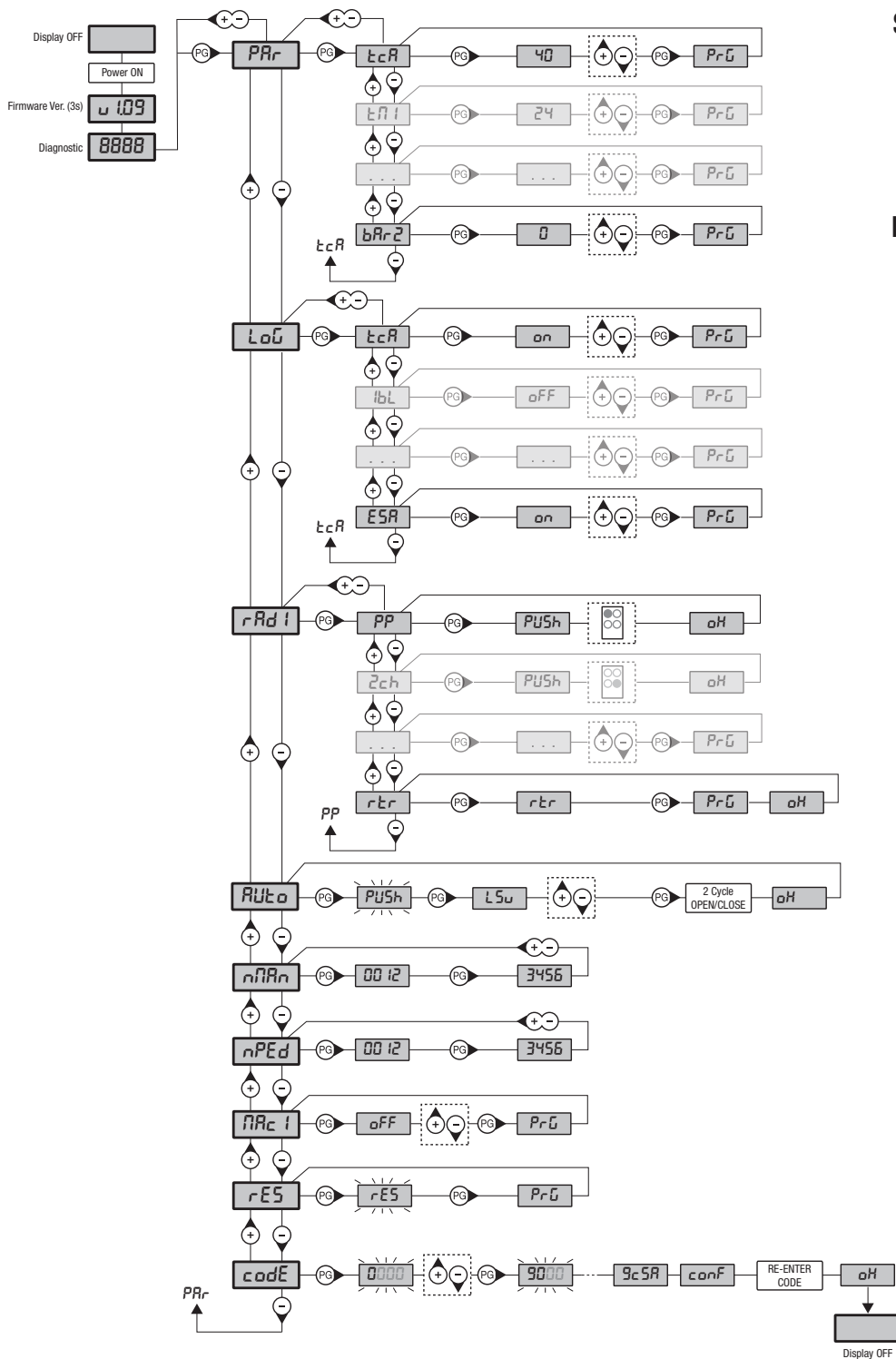
AUX3:0006

Input NC ALARM

AUX3:0007







Schema menu di programmazione

Menu programming layout

Diagramm Programmiermenü

Menu de programmation

Menú de la carta de programación

Układ menu programowania

Legenda

	Premere il tasto (-) / Press key (-) / Die Taste (-) drücken / Appuyez sur la touche (-) / Presionar la tecla (-) / Wcisnąć przycisk (-)
	Premere il tasto (+) / Press key (+) / Die Taste (+) drücken / Appuyez sur la touche (+) / Presionar la tecla (+) / Wcisnąć przycisk (+)
	Premere il tasto (PG) / Press key (PG) / Die Taste (PG) drücken / Appuyez sur la touche (PG) / Presionar la tecla (PG) / Wcisnąć przycisk (PG)
	Premere simultaneamente (+) e (-) / Press simultaneously keys (+) and (-) / Gleichzeitig (+) und (-) drücken Presser simultanément (+) et (-) / Presionar simultáneamente (+) y (-) / Naciśnąć jednocześnie (+) i (-)
	Selezionare il valore desiderato con i pulsanti (+) e (-) / Increase/decrease the value with keys (+) and (-) Mit den Tasten (+) und (-) kann man eingerichtete Werte ändern / Régler la valeur désirée avec les touches (+) et (-) Establecer con las teclas (+) y (-) el valor deseado / Nastawia przyciskami (+) i (-) obraną wartość
	Selezionare il pulsante del trasmettitore da associare alla funzione / Press the transmitter key, which is to be assigned to function Taste des Sendegeräts drücken, dem diese Funktion zugeteilt werden soll. Appuyer sur la touche du transmetteur qu'e l'on désire affecter à cette fonction. Presionar la tecla del transmisor que se desea asignar a esta función. / Wcisnąć przycisk nadajnika, który zamierza się skojarzyć z tą funkcją.



The product shall not be used for purposes or in ways other than those for which the product is intended for and as described in this manual. Incorrect uses can damage the product and cause injuries and damages.
The company shall not be deemed responsible for the non-compliance with a good manufacture technique of gates as well as for any deformation, which might occur during use.
Keep this manual for further use.



This manual has been especially written to be use by qualified fitters.
Installation must be carried out by qualified personnel (professional installer, according to EN 12635), in compliance with Good Practice and current code.
Make sure that the structure of the gate is suitable for automation.
The installer must supply all information on the automatic, manual and emergency operation of the automatic system and supply the end user with instructions for use.



Packaging must be kept out of reach of children, as it can be hazardous. For disposal, packaging must be divided the various types of waste (e.g. carton board, polystyrene) in compliance with regulations in force.
Do not allow children to play with the fixed control devices of the product. Keep the remote controls out of reach of children.
This product is not to be used by persons (including children) with reduced physical, sensory or mental capacity, or who are unfamiliar with such equipment, unless under the supervision of or following training by persons responsible for their safety.
Apply all safety devices (photocells, safety edges, etc.) required to keep the area free of impact, crushing, dragging and shearing hazard. Bear in mind the standards and directives in force, Good Practice criteria, intended use, the installation environment, the operating logic of the system and forces generated by the automated system.
Installation must be carried out using safety devices and controls that meet standards EN 12978 and EN 12453.
Only use original accessories and spare parts, use of non-original spare parts will cause the warranty planned to cover the products to become null and void.
All the mechanical and electrical parts composing automation must meet the requirements of the standards in force and outlined by CE marking.



An omnipolar switch/section switch with remote contact opening equal to, or higher than 3mm must be provided on the power supply mains.
Make sure that before wiring an adequate differential switch and an overcurrent protection is provided.
Pursuant to safety regulations in force, some types of installation require that the gate connection be earthed.
During installation, maintenance and repair, cut off power supply before accessing to live parts.
Also disconnect buffer batteries, if any are connected.
The electrical installation and the operating logic must comply with the regulations in force.
The leads fed with different voltages must be physically separate, or they must be suitably insulated with additional insulation of at least 1 mm.
The leads must be secured with an additional fixture near the terminals.
During installation, maintenance and repair, interrupt the power supply before opening the lid to access the electrical parts
Check all the connections again before switching on the power.
The unused N.C. inputs must be bridged.



WASTE DISPOSAL

As indicated by the symbol shown, it is forbidden to dispose this product as normal urban waste as some parts might be harmful for environment and human health, if they are disposed of incorrectly. Therefore, the device should be disposed in special collection platforms or given back to the reseller if a new and similar device is purchased. An incorrect disposal of the device will result in fines applied to the user, as provided for by regulations in force.

Descriptions and figures in this manual are not binding. While leaving the essential characteristics of the product unchanged, the manufacturer reserves the right to modify the same under the technical, design or commercial point of view without necessarily update this manual.

TECHNICAL DATA

Mains power supply	230 Vac $\pm 10\%$ 50/60 Hz (120 Vac $\pm 10\%$ 50/60 Hz for BRAINY PLUS 115)
Consumption in Stand-by	0.6 W*
Output, Motor	1/2 motor, 230 Vac $\pm 10\%$ (1/2 motors 120 Vac $\pm 10\%$ for BRAINY PLUS 115)
Motor maximum power	600W + 600W
Output, power supply of accessories	24Vdc 1A max.
Protection level	IP54
Operating temperature	-20°C / +60°C
Radio receiver	433,92 MHz, incorporated and configurable
No. of codes storable in memory	512 (2048 with MEM2048 module)

* The control unit enters energy-saving mode (standby) only after the completion of any timed functions that may have been activated (TCA, courtesy light, etc.), but in any case no later than 240 seconds after the end of the maneuver. To activate standby mode with the gate open (TCA=OFF), it is necessary to disable the signaling functions (e.g., SCA).

BRAINY PLUS CONTROL UNIT

INPUT/OUTPUT FUNCTIONS

N° Terminals	Function	Description
1-2-3	Power supply	BRAINY PLUS: Input 230 Vac $\pm 10\%$ 50/60Hz (1-GND/2-Phase/3-Neutral) BRAINY PLUS 115: Input 120 Vac $\pm 10\%$ 50/60Hz (1-GND/2-Phase/3-Neutral)
4-5-6	Motor 1	Connection of motor 1: (4-start/5-Com/6-start)
7-8-9	Motor 2	Connection of motor 2: (7-start/8-Com/9-start)
10-11	Blinking light	Connection of blinking light 230Vac 40W max.
12-13	AUX3	N.O. clean contact VOLTAGE FREE (230V/16a max) for courtesy light, 2° CH radio, etc. The operating mode can be changed using parameter AUX3.
14-15	24 Vdc	Power supply output accessories 24Vdc/1A max.
16-17	12/24Vdc lock	Power supply output accessories 12/24Vdc/10Wmax for electric lock. The VLOC parameter selects the output voltage, the MLOC logic the type of electric lock.
18-19	AUX1	Normally Open clean contact (24Vac/1A max). The operating mode can be changed via parameter AUX1.
20-21	Sensitive Edge 2	Sensitive Edge 2 contact input. See BAR2 parameter for configuration of the sensitive edge type and operating mode. The Sensitive Edge stops movement of the gate and inverts it for approximately 3s.
22-23	Sensitive Edge 1	Sensitive Edge 1 contact input. See BAR1 parameter for configuration of the sensitive edge type and operating mode. The Sensitive Edge stops movement of the gate and inverts it for approximately 3s.
24	SWO-M1	OPEN limit switch input motor 1 (N.C contact.)
25	SWC-M1	CLOSE limit switch input motor 1 (N.C. contact)
26	SWO-M2	OPEN limit switch input motor 2 (N.C. contact)
27	SWC-M2	CLOSE limit switch input motor 2 (N.C. contact)
28-29	COM	Common for limit switch and all the control inputs.
30	Step-by-step (IN1)	Step-by-step button input (N.O. contact). Configurable via SS logic and the advanced functions IN 1
31	PED (IN2)	Pedestrian button input (N.O. contact) Configurable via the TPED parameter and the advanced functions IN 2
32	OPEN (IN3)	OPEN button input (N.O. contact). You can connect a timer for opening in time slots. Configurable via the advanced functions IN 3
33	CLOSE (IN4)	CLOSE button input (N.O. contact). Configurable via the advanced functions IN 4
34	PHOT OPEN (IN5)	Photocell input enabled in opening and closure. Using the auxiliary outputs AUX1/2/3/4, you can connect the photocells in verified mode. See Fig. 2/3. Other uses of the input are possible through the advanced functions IN 5
35	PHOT CLOSE (IN6)	Photocell input active only in closure. Using the auxiliary outputs AUX1/2/3/4, you can connect the photocells in verified mode. See Fig. 2/3. Other uses of the input are possible through the advanced functions IN 6
36	STOP	STOP button input (N.C. contact)
37-38	Antenna	Antenna connection for plug-in radio receiver board (35-screen/36-signal).
41-42	AUX2	Voltage-free Normally Open Contact (24Vac/1A max). The operating mode can be changed via parameter AUX2.
43-44	AUX4	Voltage free N.O. contact (24Vac/1A max). The operating mode can be changed using parameter AUX4.
EXP 1	Expansion port 1	Expansion connector for serial KNX or pro.UP
EXP 2	Expansion port 2	Expansion connector for serial KNX or pro.UP
EXP 3	Expansion port 3	Expansion connector for optional cards type SIS
EXP 4	Expansion port 4	Expansion connector for optional cards type SIS
EXP 5	Expansion port 5	Expansion connector for optional cards type SIS

The control unit is equipped with an built-in radio module for the reception of variable code controls, with ARC (Advanced Rolling-Code) or fixed code, 433.92 MHz frequency.

TO CHECK CONNECTIONS:

- 1) Cut-off power supply.
- 2) Manually release the wings, move them to approx. half-stroke and lock them again.
- 3) Reset power supply.
- 4) Send a step-by-step control signal by pressing the button or the remote control key.
- 5) The wings should start an opening movement.
If this is not the case, invert the movement wires of the motor. (4<->6 for motor M1, and 7<->9 for motor M2) and the relevant limit switch inputs (22<->23 for motor M1, and 24<->25 for motor M2).

AUTOSET

This function is used to set the optimal automation operating values and, at the end of the procedure, the WORKING TIME is set. For motors with Encoder, the applied torque values are also set.

To autose, proceed as follows:

- 1) Ensure that in the manoeuvre area of the gates there are no obstacles of any kind.
If necessary, block off the area to prevent access to people, animals, cars, etc.
During the autose phase, the anti-crushing function is not active.
- 2) Press the PG button to access programming, with the button + select the AUTO function and press PG.
- 3) Select the type of motor used:
NLSU Motors without limit switch and encoder.
ENC Motors with Encoder
LSU Motors with electromechanical limit switch.
- 4) Press OK to start the autose phase.
- 5) If NLSU has been selected, you must start the AUTOSET with the doors CLOSING and set the limit switches with the PG key, as indicated below.
The control unit commands an opening manoeuvre of motor 1, the display shows OPM1, once the desired stop point has been reached, press the PG button
The control unit commands an opening manoeuvre of motor 2, the display shows OPM2, once the desired stop point has been reached, press the PG button
The control unit commands a closing manoeuvre of the motor 2, the display shows CLM2, once the desired stop point has been reached, press the PG button
The control unit commands a closing manoeuvre of the motor 1, the display shows CLM1, once the desired stop point has been reached, press the PG button
Then, the control unit automatically controls a complete opening and closing cycle at the end of which the OK message is displayed and the AUTOSET phase has ended.
- 6) If ENC/LSU has been selected, the central unit performs a sequence of operations: single partial openings, full openings and closings at different speeds, and so on.
During this phase, the display will show some acronyms that indicate the operation that is being performed at that time:
OPM1/2: motor 1 or 2 in opening phase
CLM1/2: motor 1 or 2 in closing phase.
If the motor movement is opposite to display indications, stop the autose by pressing any of the programming buttons, reverse the +/- wires of the motor and repeat the autose operation. 6) When the autose phase has ended, the message OK is displayed.

Notes:

If the autose is not successful, an ERR error message is displayed, refer to the Error Message table and proceed accordingly, and then repeat the autose operation.

ARC COMPATIBLE CONTROL UNIT

IMPORTANT, PLEASE READ CAREFULLY:

The radio receiver in this product is compatible with the new ARC (Advanced Rolling Code) transmitters which, thanks to 128-bit encryption ensure superior copy-security.

Memorisation of new ARC transmitters is very similar to that of normal Rolling Code transmitters with HCS* code, but you should note the following:

- 1) ARC transmitters and Rolling Code HCS cannot be memorised in a single receiver.
- 2) The first transmitter memorised determines the type of transmitters to be used later. If the first transmitter memorised is ARC, you cannot store Rolling Code HCS transmitters, and vice versa.
- 3) Fixed code transmitters may only be used in conjunction with Rolling code HCS transmitters, bringing the-CVAR logic off. Therefore, they cannot be used combined with ARC transmitters. If the first Rolling Code transmitter memorised is an ARC CVAR, the logic is inoperative.
- 4) If you want to change the type of transmitters it is necessary to proceed with a receiver reset.

PROGRAMMING

The programming of the various functions of the control unit is carried out using the LCD display on the control unit and setting the desired values in the programming menus described below.

The parameters menu allows you to assign a numerical value to a function, in the same way as a regulating trimmer.

The logic menu allows you to activate or deactivate a function, in the same way as setting a dip-switch.

Other special functions follow the parameters and logic menus and may vary depending on the type of control unit or the software release.

TO ACCESS PROGRAMMING

- 1 - Press the button <PG>, the display goes to the first menu, Parameters "PAR".
- 2 - With the <+> or <-> button, select the menu you want.
- 3- Press the button <PG>, the display shows the first function available on the menu.
- 4 - With the <+> or <-> button, select the function you want.
- 5 - Press the button <PG>, the display shows the value currently set for the function selected.
- 6 - With the <+> or <-> button, select the value you intend to assign to the function.
- 7 - Press the button <PG>, the display shows the signal "PRG" which indicates that programming has been completed.

NOTES

Pressing <-> with the display turned off means an impulse of P.P.

Simultaneously pressing <+> and <-> from inside a function menu allows you to return to the previous menu without making any changes.

Hold down the <+> key or the <-> key to accelerate the increase/decrease of the values.

After waiting 30s the control unit quits programming mode and switches off the display.

PARAMETERS, LOGIC AND SPECIAL FUNCTIONS

The tables below describe the individual functions available in the control unit.

PARAMETERS (PRr)			
MENU	FUNCTION	MIN-MAX- (Default)	MEMO
tcR	Automatic closing time. Active only with logic "TCA"=ON. At the end of the set time the control unit orders a closing manoeuvre.	1-240-(40s)	
tm1	Motor 1 work time. Adjusts the maximum duration of the motor 1 opening and closing manoeuvre. The value is automatically set by the AUTOSET function. You can still edit it later.	5-180-(24s)	
tm2	Motor 2 work time. Adjusts the maximum duration of the motor 2 opening and closing manoeuvre. The value is automatically set by the AUTOSET function. You can still edit it later.	5-180-(24s)	
tPEd	The operating time for partial opening (pedestrian) is adjusted on motor 1. With ENC: ON, the value is the opening percentage over the total stroke memorised during self-setting. With ENC: OFF, the value is the opening percentage over the time preset in TM1 parameter.	1-99-(50%)	
Pno1	The torque applied to motor 1 in the OPENING* phase is adjusted.	1-99-(40%)	
Pnc1	The torque applied to motor 1 in the CLOSING* phase is adjusted.	1-99-(40%)	
Pno2	The torque applied to motor 2 in the OPENING* phase is adjusted.	1-99-(40%)	
Pnc2	The torque applied to motor 2 in the CLOSING* phase is adjusted.	1-99-(40%)	
PSO1	It adjusts the torque applied during the slowing phase in the OPENING of the motor 1*.	1-99-(70%)	
PSc1	It adjusts the torque applied during the slowing phase during the CLOSING of the motor 1*.	1-99-(70%)	
PSO2	It adjusts the torque applied during the slowing phase in the OPENING of the motor 2*.	1-99-(70%)	
PSc2	It adjusts the torque applied during the slowing phase during the CLOSING of the motor 2*.	1-99-(70%)	
tdno	Mot.2 opening delay time. Regulates the delay time of motor 2 on opening with respect to motor 1	0-15-(2s)	
tdnc	Mot.1 closing delay time Regulates the delay time of motor 1 on closing with respect to motor 2	0-60-(3s)	
tsn1	Motor 1 slowdown It adjusts the duration of the slowing phase of the motor 1. The value is expressed as a percentage of the total value of the stroke. With 0 value, slowdown is bypassed.	0-99-(0%)	
tsn2	Motor 2 slowdown It adjusts the duration of the slowing phase of the motor 2. The value is expressed as a percentage of the total value of the stroke. With 0 value, slowdown is bypassed.	0-99-(0%)	
tlS	TLS contact activation time (only if one of the AUX parameters is set to 2). With each operation, the contact (AUX1/AUX2/AUX3/AUX4) closes for the set time.	1-240-(60s)	
t2ch	Sets the switching time of the second radio channel. 0:bistable output, on each activation of the second radio channel the output changes status from 1 to 1250: switching time in seconds	0-250-(1)	
RLrN	Alarm output enables when at least one of the following inputs remains active for the time set (STOP - PHOT - PHOTC - BAR - SWO+SWC). It is necessary for one of the parameters AUX 1 is set to 7 (Alarm output) Value expressed in seconds.	10-240 (60)	
tlOc	Electric lock activation time. If the electric lock is not used, set the parameter at 0.	0-5-(3s)	
uLoc	Electric lock power supply voltage. 0: 12V - 1:24V	0-1-(0)	

SEAU	The intervention threshold of the anti-crashing device (Encoder) during the phase at normal speed is adjusted.* 0: Off - 1: minimum sensitivity - 99: maximum sensitivity	0-99-(0%)	
SEAr	The intervention threshold of the anti-crashing device (Encoder) during braking is adjusted *. 0: Off - 1: minimum sensitivity - 99: maximum sensitivity	0-99-(0%)	
t Inc	This parameter is enabled only for motors equipped with Encoder. The encoder inhibition is regulated near the opening and closing mechanical stoppers. 1: minimum distance – 250: maximum distance	1-250-(250)	
bLco	It adjusts the stoppage time after opening limit switch interception. Recommended for motors for swing doors equipped with limit switch. Value expressed in tenths of a second	0-50-(0)	
bLcc	It adjusts the stoppage time after closing limit switch interception. Recommended for motors for swing doors equipped with limit switch. Value expressed in tenths of a second	0-60-(0)	
AUX 1	Sets the operating mode of the AUX1 output (N.O. contact) 0: SCA output (gate open indicator light). The light is off when the door is closed, flashes when the door is moving, is on when the door is open. 1: 2CH radio output. The output is controlled by the second radio channel of the built-in receiver (see RADIO menu) 2: courtesy light output (activation time is set by parameter TLS) 3: Zone light output. The contact closes for the duration of the manoeuvre and for the duration of the TCA, it only reopens when the gate is closed. 4: accessory power supply output (for .photocell verification - coast, in combination with logics TST1-TST2-TST3) 5: flashing output 6: gate open alarm output (gate open for twice the set TCA time) 7: NC input alarm output or card error (NC=alarm not active, NO=alarm active)	0-7-(0)	
AUX2	Same settings as parameter AUX1 but referring to output AUX2 (N.O contact)	0-7-(1)	
AUX3	Same AUX1 parameter settings but referring to AUX3 output (N.O. contact) ATTENTION: The AUX3 output is a N.O. clean contact VOLTAGE FREE (230V/16a max)	0-7-(2)	
AUX4	Same AUX1 parameter settings but referring to AUX4 output (N.O. contact)	0-7-(3)	
bAr 1	Sets the operating mode of the BAR 1 input 0: NC Sensitive Edge Input. Active in Opening and Closing 1: NC Sensitive Edge Input Verified. Active in Opening and Closing 2: Sensitive Edge Input 8k2. Active in Opening and Closing More configuration options are available among the advanced functions	0-2-(0)	
bAr 2	Sets the operating mode of the BAR 2 input 0: NC Sensitive Edge Input. Active in Opening and Closing 1: NC Sensitive Edge Input Verified. Active in Opening and Closing 2: Sensitive Edge 8k2 Input. Active in Opening and Closing More configuration options are available among the advanced functions	0-2-(0)	
** WARNING: AN INCORRECT SETTING OF THESE PARAMETERS MAY RESULT IN A DANGER. COMPLY WITH REGULATIONS IN FORCE! For hydraulic motors, set the value to maximum (99). Use by-pass valves to adjust the applied torque.			

LOGIC (L o G)			
MENU	FUNCTION	ON-OFF-(Default)	MEMO
t c R	Enables or disables automatic closing On: automatic closing enabled Off: automatic closing disabled	(ON)	
ibL	Enables or disables condominium function. Off: condominium function disabled. On: condominium function enabled. The step-by-step impulse or transmitter impulse has no effect during the opening phase.	(OFF)	
ibc R	During the TCA phase, the PP controls are enabled or disabled. On: PP controls are disabled. Off: PP controls are enabled.	(OFF)	
ScL	Enables or disables rapid closing On: rapid closing enabled. With the gate open or in the opening phase the intervention of the photocell causes automatic closing after 3 s. Active only with TCA:ON. Off: rapid closing disabled.	(OFF)	
PP	Selects the operating mode of the "Step by step button" and of the transmitter. On: Operation: OPEN > CLOSE > OPEN > Off: Operation: OPEN > STOP > CLOSE > STOP >	(OFF)	
PrE	Enables or disables pre-blinking. On: Pre-blinking enabled. Blinking is activated 3s before the motor starts. Off: Pre-blinking disabled.	(OFF)	

hRn	Enables or disables the inversion stroke function On: Function enabled. Before each opening manoeuvre the control unit orders a manoeuvre of 2s in the opposite direction to facilitate the release of the electric lock. Off: Function disabled.	(OFF)	
bLc	Enables or disables the block maintaining function. Recommended for hydraulic motors to keep the leaf resting against the mechanical stop block. On: Block maintaining function enabled. Every 2 hours the control unit makes a closing manoeuvre with a duration of about 3s to keep the leaf in contact. Off: Block maintaining function disabled.	(OFF)	
SPn	Enables or disables starting torque function. On: Starting torque enabled. At the start of each manoeuvre for 2s the motor operates at maximum torque. Off: Starting torque disabled.	(ON)	
LtAR	Selects the operating mode of the blinking light during the time TCA On: Blinking light on during TCA Off: Blinking light off during TCA	(OFF)	
htr	Enables or disables Man present function. On: Man Present operation. The OPEN/CLOSE buttons must be held down during the whole manoeuvre. Off: Automatic operation.	(OFF)	
iNoE	Select the 1/2 motors operating mode: On: only one motor (motor 1) active. Function to be used in the following cases: - for single motor, to connect M1:4-5-6. - for two synchronized motors (for instance overhead door), to connect M1:4-5-6 and M2:7-8-9. You need to regulate the parameters related to the motor 1, the limit switch entries M2 are disarmed (not activate). Off: Both motors operating.	(OFF)	
nBLH	It sets the BLINK flashing output operating mode. On: The output provides intermittent voltage during the manoeuvre phase (2 flashes per second). Off: The output provides continuous voltage during the manoeuvre phase.	OFF	
noE	Enables or disables the calculation of the residual working time in the case of partial manoeuvres: On: Time calculation disabled. In the event of partial manoeuvres, the working time is reset to zero. The next manoeuvre resumes for the entire duration set by the TM1/TM2 parameter. Off: Time calculation enabled. In the event of partial manoeuvres, the working time is memorised and subtracted from the value of the parameter TM1/TM2 in the next manoeuvre.	ON	
nInu	Motors start order: On: motor 2 starts first in opening. Off: motor 1 starts first in opening.	OFF	
noE	The calculation of residual operating time is activated or deactivated in the event of partial operations: On: Calculation of deactivated time. In case of partial operations, the operating time is reset. The following operation restarts for the entire time preset by parameter TM1/TM2. Off: Calculation of activated time. In case of partial operations, the operating time is stored in memory and then subtracted from the TM1/TM2 parameter value in the following operation.	(ON)	
cuAr	Enables or disables the programmable code transmitters. IMPORTANT: The transmitters with programmable code can only be used in conjunction with the Rolling Code HCS transmitters. On: Radio receiver enabled only for transmitters Rolling Code (ARC or HCS, the first transmitter sets the operating mode). Off: Receiver enabled for transmitters Rolling Code HCS and programmable (self-learning and dip/switch).	(ON)	
nLoc	Selects the type of electric lock used. On: Magnetic electric lock, normally fed at 12Vdc. Before each opening manoeuvre the power supply is interrupted for the time set by the parameter TLOC. Note: When MLOC is set to ON, energy saving is automatically disabled (ESA:OFF) as it is not possible to keep the electric lock powered in stand-by mode. Off: Electric lock with latch, normally not fed. Before each opening manoeuvre power is fed at 12Vdc for the time set by the parameter TLOC.	(OFF)	
bb	The thrust function in the closing phase is activated or deactivated by this logic. On: the operation in the closing phase is carried out at normal speed during the last second (braking is disabled). In this way, a better hooking of the electric lock is performed. Off: disabled function.	(OFF)	
horN	The automatic switching is activated/deactivated in Service Man mode. On: If the sensitive edge remains activated for at least 10 seconds with open or closed gate, the control unit automatically switches to Service man mode. Off: Disabled function.	(OFF)	

CuAr	The CVAR Logic function depends on the operating mode of the receiver: ARC configured receiver (the first transmitter memorised is ARC): On: The AK series transmitters closed from an ARC transmitter already memorised are enabled. Off: Disable reception of AK clonable transmitters HCS configured receiver (the first transmitter memorised is HCS): On: Enables reception of programmable transmitters (self-learning and dip/switch) Off: Disables reception of programmable transmitters (self-learning and dip/switch).	(ON)	
rEn	The remote storage of the radio transmitter codes is enabled or disabled (see par. REMOTE LEARNING). On: Enabled remote storage Off: Disabled remote storage.	(ON)	
ESR	It enables or disables the "ESA" energy saving functionality. On: The control unit, once the manoeuvre has terminated and enabling time is up of the service light, removes the power supply to the accessories output, setting in stand-by status. Off: Energy saving disabled. To use if you want to have the accessories power supply always enabled, for example if using 24 Vdc, or other devices that need to be always powered. Note: If the ESA logic is set to ON, prior to performing each manoeuvre, the unit checks the switching of the photocell contact. If the check fails, the operation does not start.	(ON)	

RADIO (rAd)

MENU	FUNCTION
PP	By selecting this function, the receiver is waiting for (Push) a transmitter code to be assigned to the step-by-step function. Press the transmitter key, which is to be assigned to this function. If the code is valid, it will be stored in memory and OK will be displayed. If the code is not valid, the Err message will be displayed.
2ch	By selecting this function, the receiver is waiting for (Push) a transmitter code to be assigned to the second radio channel. Press the transmitter key, which is to be assigned to this function. If the code is valid, it will be stored in memory and OK will be displayed. If the code is not valid, the Err message will be displayed.
PEd	When this function is selected, the receiver awaits (Push) a transmitter code to be assigned to the pedestrian opening function (see TPED parameter). Press the transmitter key, which is to be assigned to this function. If the code is valid, it will be stored in memory and OK will be displayed. If the code is not valid, the Err message will be displayed.
Stop	Selecting this function, the receiver sets in standby (Push) of a transmitter code to assign to the STOP function. Press the transmitter key you intend to assign to this function. If the code is valid, it is saved and the message OK is displayed If the code is not valid, the message Err is displayed
nH	Selecting this function, the LCD display shows the number of transmitters currently saved in the receiver.
clr	By selecting this function, the receiver is waiting for (Push) a transmitter code to be erased from memory. If the code is valid, it will be stored in memory and OK will be displayed. If the code is not valid, the Err message will be displayed.
etr	The memory of the receiver is entirely erased. Confirmation for the operation is asked. By selecting this function, the receiver waits for (Push) the GPM key to be pressed again to confirm the operation. At end of erasing, the OK message is displayed

Note: The transmitters are stored in an EPROM memory (Fig.1 -U11), which can be removed and repositioned in a new control unit, it required.

CYCLES NUMBER (nAn)

Displays the number of complete cycles (open+close) carried out by the automation.

When the <PG> button is pressed for the first time, it displays the first 4 figures, the second time it shows the last 4. Example <PG> 00 12 >>> <PG> 3456: made 123.456 cycles.

NUMBER OF PEDESTRIAN MANOEUVRES (nPEd)

Displays the number of pedestrian manoeuvres (PED input activation) carried out by automation. The first press of the <PG> button displays the first 4 digits, the second press displays the last 4. e.g. <PG> 0012 >>> <PG> 3456: executed 123.456 cycles.

MAINTENANCE CYCLES (nAc i)

This function enables to activate the maintenance request notice after a number of manoeuvres determined by the installer.

To activate and select the number of manoeuvres, proceed as follows:

Press button <PG>, the display will show OFF, which indicated that the function is disabled (default value).

With the buttons <+> and <-> select one of the numeric values proposed (from OFF to 100). The values are intended as hundreds of cycles of manoeuvres (for example: the value 50 indicates 5000 manoeuvres).

Press the OK button to activate the function. The display will show the message *Prac*.

The maintenance request is indicated to the user by keeping the indicator lamp lit up for other 10 sec after the conclusion of the opening or closing operation.

RESET (rE5)

RESET of the control unit. ATTENTION!: Returns the control unit to the default values.

Pressing the <PG> button for the first time causes blinking of the letters rE5, pressing the <PG> button again resets the control unit. Note: The transmitters are not erased from the receiver nor is the access password.

All the logics and all the parameters are brought back to default values, it is therefore necessary to repeat the autoset procedure.

AUTOSET (AutO)

SEE AUTOSET PARAGRAPH

PROTECTION CODE (codE)

It allows to type in an access protection code to the programming of the control unit.

A four-character alphanumeric code can be typed in by using the numbers from 0 to 9 and the letters A-B-C-D-E-F.

The default value is 0000 (four zeros) and shows the absence of a protection code.

While typing in the code, this operation can be cancelled at any moment by pressing keys + and – simultaneously. Once the password is typed in, it is possible to act on the control unit by entering and exiting the programming mode for around 10 minutes in order to allow adjustments and tests on functions.

By replacing the 0000 code with any other code, the protection of the control unit is enabled, thus preventing the access to any other menu. If a protection code is to be typed in, proceed as follows:

- select the Code menu and press OK.
- the code 0000 is shown, also in the case a protection code has been previously typed in.
- the value of the flashing character can be changed with keys + and -.
- press OK to confirm the flashing character, then confirm the following one.
- after typing in the 4 characters, a confirmation message “CONF” appears.
- after a few seconds, the code 0000 appears again
- the previously stored protection code must be reconfirmed in order to avoid any accidental typing in.

If the code corresponds to the previous one, a confirmation message “oH” appears.

The control unit automatically exits the programming phase. To gain access to the Menus again, the stored protection code must be typed in.

IMPORTANT: TAKE NOTE of the protection code and KEEP IT IN A SAFE PLACE for future maintenance operations.

To remove a code from a protected control unit it is necessary to enter into programming with the password and bring the code back to the 0000 default value.

IF YOU LOOSE THE CODE, PLEASE CONTACT THE AUTHORISED SERVICE CENTER FOR THE TOTAL RESET OF THE CONTROL UNIT.

BRAKING

With the ENC=OFF logics, braking is given by the TSM1/TSM2 parameter value referred to the TM1/TM2 operating time. For example, if the operation of motor 1 lasts 20 seconds and TM1=24s and TSM1=8 are preset, 4 s braking will result (20-(24-8).

With ENC=ON logics, braking is calculated as a percentage on the space covered during the complete operation. If, for example, TSM1=20, 20% of the operation will be slow down.

FUSES

F1: Motor Protection	F6.3A (230Vac) / F10A (120Vac)
F2: Primary protection power supply	T1A -
F3: T800mA accessories power supply output protection	T800 mA

TRANSMITTER REMOTE LEARNING

If the transmitter code is already stored in the receiver, the remote radio learning can be carried out (without accessing the control unit).

IMPORTANT: The procedure should be carried out with gate in the opening phase, during the TCA dwell time.

Proceed as follows:

- 1 Press the hidden key of the transmitter, the code of which has already been stored in memory.
- 2 Within 5 seconds, press the already memorised transmitter key corresponding to the channel to be matched to the new transmitter. The flashing light switches on.
- 3 Within 10 seconds, press the hidden key of the new transmitter.
- 4 Within 5 seconds, press the key of the new transmitter to be matched to the channel selected at item 2. The flashing light switches off.
- 5 The receiver stores the new transmitter code and exits from the programming mode immediately.

EPROM 64 TX READING

If BRAINY PLUS is installed to replace an old BRAINY model, the EPROM can be read from 64 transmitters to transfer them to the new EPROM 512 installed in the BRAINY PLUS, proceed as follows:

- 1) Cut power to the BRAINY PLUS control unit
 - 2) Remove the EPROM 512 (FIG. 1 -U9)
 - 3) Insert the old EPROM 64 from BRAINY respecting the insertion direction
 - 4) Give mains power, the display shows the BKUP message, wait for the OK message, disconnect power and remove EPROM 64
 - 5) Reinsert the EPROM 512 and return power, the REST (reset) message is displayed followed by OK (or KO if reset is unsuccessful). After a few seconds, the firmware version of the control unit is displayed, the procedure is finished, the ARC or HCS transmitters have been memorised in the new EPROM.
- If you replace a BRAINY PLUS with another BRAINY PLUS, the procedure described above is not necessary, since the EPROMs are compatible, simply replace the new EPROM with the old one.

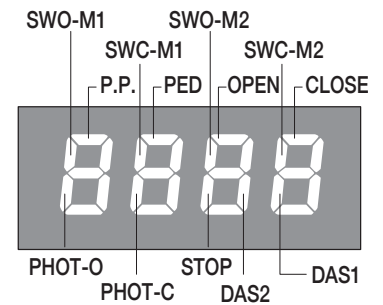
ERROR MESSAGES

Some messages that are displayed in the event of malfunctions are shown hereunder:

<i>Enc 1</i>	motor 1 encoder error	check motor 1 encoder connection
<i>Enc 2</i>	motor 2 encoder error	check motor 2 encoder connection
<i>ANP 1</i>	motor 1 obstacle error	Check the presence of obstacles on the stroke of the motor gate 1
<i>ANP 2</i>	motor 2 obstacle error	Check the presence of obstacles on the stroke of the motor gate 2
<i>Err 1</i>	motor 1 circuit test error	check motor 1 connections
<i>Err 2</i>	motor 2 circuit test error	check motor 2 connections
<i>Err 3</i>	run relay test error	request technical support
<i>Err 4</i>	PHOT photocell test error	check connections, alignment of the photocell PHOT C or obstacles present.
<i>Err 5</i>	PHOT CLOSE photocell test error	check connections, alignment of the PHOTOCELL photo close or obstacles present.
<i>Err 6</i>	Sensitive Edge error active (during autoset)	during the autoset phase, the Sensitive Safety Edge activated.
<i>Err 7</i>	active stop error (during autoset)	during the autoset phase, the STOP input activated.
<i>Err 8</i>	active input error (during autoset)	during the autoset phase, a start/Pedestrian/Open/Close input activated.
<i>ERcc</i>	accessory output error	the maximum current threshold for supplying accessories has been exceeded.
<i>ELoc</i>	lock output error	the maximum current threshold available for supplying the electric lock has been exceeded.

DIAGNOSTICS

During normal operation, the LCD display shows the status of the inputs and outputs as shown in the diagram opposite. Each activation of an input/output corresponds to the switching on of the corresponding segment of the LCD display.



ADVANCED FUNCTIONS

The advanced functions that are not normally displayed in the programming menu are described below.

To view the advanced functions it is necessary, with the display off, to press and hold the PG button for 5s.

Advanced functions now appear in the queue of the standard functions and in some cases replace them, allowing higher customising of the settings.

PARAMETRS (PR)			
MENU	FUNCTION	MIN-MAX- (Default)	MEMO
t_{n1}	Motor 1 work time in OPENING. Adjusts the maximum duration of the opening manoeuvre of motor 1. It must be set about 4 seconds longer than the actual running time of the automatism.	5-180-(24s)	
t_{n2}	Motor 1 work time in CLOSING. Adjusts the maximum duration of the closing manoeuvre of motor 1. It must be set about 4 seconds longer than the actual running time of the automatism.	5-180-(24s)	
	Replaces the t_{n1} parameter		
t_{n2}	Motor 2 work time in OPENING. Adjusts the maximum duration of the opening manoeuvre of motor 2. It must be set about 4 seconds longer than the actual running time of the automatism.	5-180-(24s)	
t_{n2}	Motor 2 work time in CLOSING. Adjusts the maximum duration of the closing manoeuvre of motor 2. It must be set about 4 seconds longer than the actual running time of the automatism.	5-180-(24s)	
	Replaces the t_{n2} parameter		
t_{s1}	Motor 1 Slowdown in OPENING Adjusts the duration of the slowdown phase in the opening of motor 1. The value is expressed as a percentage of the total value of the stroke. With 0 value, slowdown is bypassed.	0-99-(0%)	
t_{s1}	Motor 1 Slowdown in CLOSING Adjusts the duration of the slowdown phase in the closing of motor 1. The value is expressed as a percentage of the total value of the stroke. With 0 value, slowdown is bypassed.	0-99-(0%)	
	Replaces the t_{s1} parameter		
t_{s2}	Motor 2 Slowdown in OPENING Adjusts the duration of the slowdown phase in the opening of motor 2. The value is expressed as a percentage of the total value of the stroke. With 0 value, slowdown is bypassed.	0-99-(0%)	
t_{s2}	Motor 2 Slowdown in CLOSING Adjusts the duration of the slowdown phase in the closing of motor 2. The value is expressed as a percentage of the total value of the stroke. With 0 value, slowdown is bypassed.	0-99-(0%)	
	Replaces the t_{s1} parameter		
$SRNP$	It adjusts the sensitivity of the amperometric sensor.		
$PSLd$	Sets the pause time when switching from normal to slow speed. Value expressed in hundredths of a second. Default 10 corresponds to 1 tenth of a second.	0-50-(10)	
$in1$	It allows you to change the function of input 1 (S.S.) 0: Step-by-step (default) - with configurable operation from the SS Logic 1: Pedestrian - partial opening of motor 1 configurable from the PED parameter 2: Open - OPEN button function (N.O. contact) 3: Close - CLOSE button function (N.O. contact) 4: Photo Open - Input for N.C. contact for active photocells in the opening phase 5: Photo Close Input for N.C. contact for active photocells in the closing phase 6: Photo open verified - Input for N.C. contact for photocells active in the opening phase from use in PHOTO TEST MODE 7: Photo Close verified - Input for N.C. contact for active photocells in the closing phase from use in PHOTO TEST MODE 8: STOP - STOP button function (N.C. contact) 9: Timed Service Light activation 10: Bistable Service Light Activation	0 -10 (0)	
$in2$	Same functions as In 1 but referring to input 2 (PED)	0 -10 (1)	
$in3$	Same functions as In 1 but referring to input 3 (OPEN)	0 -10 (2)	
$in4$	Same functions as In 1 but referring to input 4 (CLOSE)	0 -10 (3)	
$in5$	Same functions as In 1 but referring to input 5 (PHOT open)	0 -10 (4)	

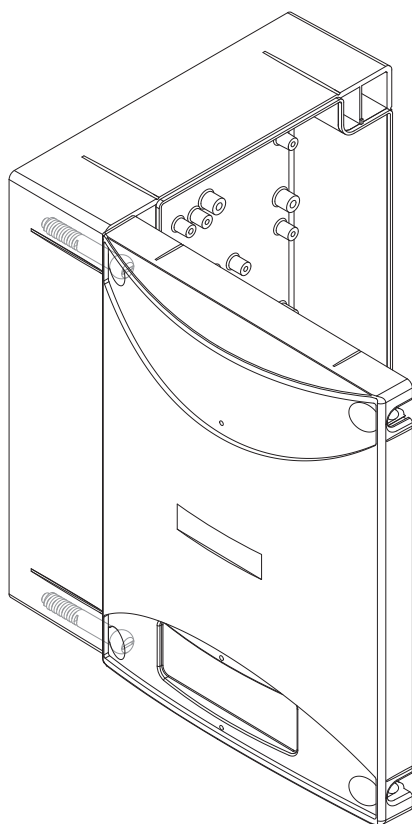
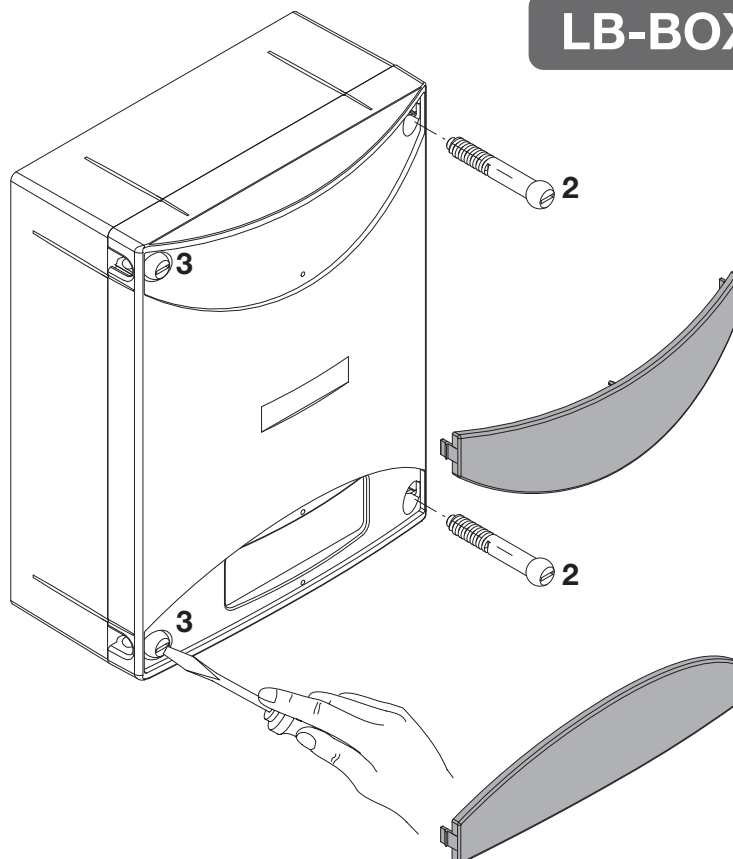
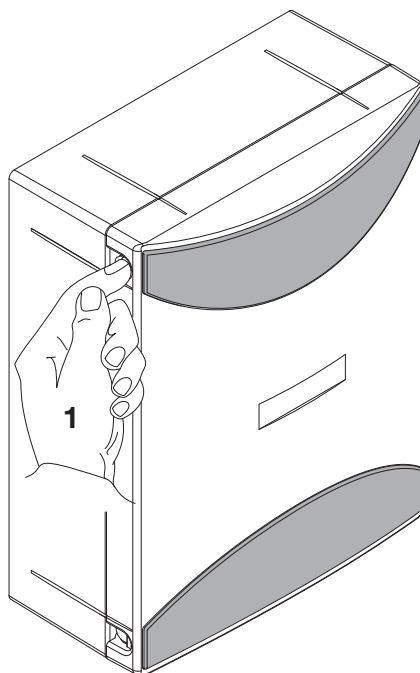
in6	Same functions as In 1 but referring to input 6 (Phot Close)	0 -10 (5)	
bar1	Sets the operating mode of the BAR 1 input 0: NC Sensitive Edge Input. Active in Opening and Closing 1: NC Sensitive Edge Input Verified. Active in Opening and Closing 2: Sensitive Edge 8k2 Input Active in Opening and Closing 3: Sensitive Edge 4k1 Input (2 8k2 Edges Parallel). Active in Opening and Closing 4: Sensitive edge input 16k4 (2 8k2 Edges in Series). Active in Opening and Closing 5: NC Sensitive Edge Input - Active in Opening 6: NC Sensitive Edge Input Verified - Active in Opening 7: Sensitive Edge Input 8k2 - Active in Opening 8: Sensitive Edge Input 4k1 (Two 8k2 Edges Parallel) - Active in Opening 9: Sensitive Edge Input 16k4 (Two 8k2 Edges in Series) - Active in Opening 10: NC Sensitive Edge Input - Active in Closing 11: NC Sensitive Edge Input Verified - Active in Closing 12: Sensitive Edge Input 8k2 - Active in Closing 13: Sensitive Edge Input 4k1 (Two 8k2 Edges Parallel) - Active in Closing 14: Sensitive Edge Input 16k4 (Two 8k2 Edges in Series) - Active in Closing	0-14 (0)	
bar2	Same functions as BAR1 but referring to input BAR2	0-14 (0)	
tdbc	Command inhibition. After receiving a command, subsequent commands are ignored for the time set. Value expressed in tenths of a second	0-50-(0s)	
t5cl	Rapid closing time	0-20 (3)	

LOGIC (L o ū)

MENU	FUNCTION	(Default)	MEMO
iblc	It enables or disables the apartment complex function in closure. On: apartment complex function enabled. The S.S. or transmitter impulse has no effect during the opening phase. Off: apartment complex function disabled.	(OFF)	
d5no	Disables N.O. commands	(OFF)	
chtr	It enables or disables the Person present function in closure. On: Person Present Operation Operation of OPEN button is impulsive, while the closure manoeuvre only takes place maintaining the CLOSE (Person present) button pressed. Off: Automatic operation.	(OFF)	

RADIO (r Rd)

MENU	FUNCTION
ch1	Selecting this function, the receiver sets in standby (Push) of a transmitter code to assign RADIO CHANNEL 1. Press the transmitter key you intend to assign to this function. If the code is valid, it is saved and the message OK is displayed If the code is not valid, the message Err is displayed
ch2	Selecting this function, the receiver sets in standby (Push) of a transmitter code to assign RADIO CHANNEL 2. Press the transmitter key you intend to assign to this function. If the code is valid, it is saved and the message OK is displayed If the code is not valid, the message Err is displayed
ch3	Selecting this function, the receiver sets in standby (Push) of a transmitter code to assign RADIO CHANNEL 3. Press the transmitter key you intend to assign to this function. If the code is valid, it is saved and the message OK is displayed If the code is not valid, the message Err is displayed
ch4	Selecting this function, the receiver sets in standby (Push) of a transmitter code to assign RADIO CHANNEL 4. Press the transmitter key you intend to assign to this function. If the code is valid, it is saved and the message OK is displayed If the code is not valid, the message Err is displayed
nch1	It sets the operating mode of radio channel 1: SS Step-by-step function 2CH: Function Second Radio Channel PED: Pedestrian function STOP: STOP Function OPEN: OPEN FUNCTION CLOS: CLOSE Function LSUT: Activate the Timed Courtesy Light according to the setting of the TLS parameter LSUB: Activates the courtesy light in bistable mode (ON>>OFF>>ON) OPM: OPEN MAINTAINED function. The opening phase is controlled by the duration the button is pressed CLM: CLOSE MAINTAINED function. The closure phase is controlled by the duration the button is pressed
nch2	Same functions as MCH1 but referring to the CH2 radio channel
nch3	Same functions as MCH1 but referring to the CH3 radio channel
nch4	Same functions as MCH1 but referring to the CH4 radio channel



1. Premere le alette sui fianchi per sganciare le due maschere copriviti.
2. Rimuovere le due viti sul lato di apertura desiderato.
3. Allentare le viti con funzione di cerniera senza rimuoverle, in modo da consentire l'apertura del coperchio.

1. Press the tabs on the sides to release the two masks that cover the screws.
2. Remove the two screws on the desired opening side.
3. Slacken the two screws that act as a hinge without removing them, so as to allow opening the cover.

1. Auf die seitlichen Laschen drücken, so dass die beiden Schraubenblenden befreit werden.
2. Die beiden Schrauben an der gewünschten Öffnungsseite ausbauen.
3. Zuletzt die beiden als Scharnier dienenden Schrauben lockern, aber nicht ausbauen, damit der Deckel geöffnet werden kann.

1. Presser les deux ailettes latérales pour décrocher les deux cache-vis.
2. Enlever les deux vis sur le côté d'ouverture désiré.
3. Desserrer les deux vis faisant fonction de charnière sans les enlever, de manière à permettre l'ouverture du couvercle.

1. Presionar las aletas en los lados para desenganchar las dos tapas cubretornillos.
2. Extraer los dos tornillos del lado de apertura deseado.
3. Afojar los dos tornillos con función de bisagra sin extraerlos, a fin de poder abrir la tapa.

1. Nacisnąć boczne klapki w celu odhaczenia dwóch masek nakrywających śruby.
2. Wyciągnąć dwie śruby po wybranej do otwierania stronie.
3. Poluzować dwie śruby blokujące bez wyciągania ich, w sposób umożliwiający otwarcie nakrywki.

BENINCA